

Applied Agentic AI for PMs/TPMs

## Capstone Project Selection Guide

*Choose the project that fits your role, interests, and daily work*

## Overview

You will complete one capstone project over the next 2 weeks. There are four projects to choose from. Each involves building an AI-powered multi-agent system, evaluating it, and documenting your design decisions. All four use the same core skills from the course but differ in domain and data.

Pick the project closest to your daily work. You will be more motivated, your design decisions will be more informed, and your writeup will be stronger if you are solving a problem you actually care about.

### **You are free to enhance:**

Once your core capabilities are working and tested, you can add features, integrate additional APIs, or bring your own use cases. The rubric rewards creativity beyond the baseline.

## The Four Projects at a Glance

|                | CalendarMate                                         | SalesGenie                                          | PRD Genie                                             | Mira                                                  |
|----------------|------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| Company        | ServionIQ (SaaS)                                     | Oak & Ember (Furniture)                             | NeuronForge (AI Startup)                              | Nexora (IT Services)                                  |
| Core Problem   | 2 hrs/day on meeting logistics                       | 2-3 hrs/day reading inquiry emails                  | Hours translating transcripts to PRDs                 | 3-4 hrs creating project plans                        |
| What You Build | Productivity assistant (briefing, scheduling, email) | Sales assistant (qualify leads, recommend products) | Documentation assistant (extract reqs, generate PRDs) | Project intelligence assistant (plans, risks, status) |
| Key APIs       | Google Calendar, Gmail (OAuth2 required)             | File upload only (no OAuth)                         | File upload only (no OAuth)                           | File/CSV upload, optional Trello API                  |
| Pattern Fit    | Router (varied request types)                        | Sequential Pipeline (same flow per email)           | Sequential Pipeline (extract → PRD → stories)         | Router (plan vs risk vs status)                       |
| Biggest Risk   | Inventing meetings                                   | Recommending products not in catalog                | Inventing requirements not in transcript              | Fabricating milestones or task statuses               |

## Tools and Concepts by Project

Use this table to choose based on the AI/Agentic AI concepts and tools you want to practice.

| Concept / Tool               | CalendarMate                                                         | SalesGenie                                                          | PRD Genie                                                                    | Mira                                                                                  |
|------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Multi-Agent Design           | 3-5 agents (orchestrator, briefing, scheduler, email, follow-up)     | 3-5 agents (ingestion, qualifier, recommender, drafter, aggregator) | 3-5 agents (extractor, PRD generator, story breakdown, gap analyzer)         | 3-5 agents (orchestrator, planner, risk assessor, status reporter, milestone tracker) |
| Orchestration Pattern        | Router pattern (varied request types)                                | Sequential Pipeline (fixed flow)                                    | Sequential Pipeline (fixed flow)                                             | Router pattern (varied request types)                                                 |
| Prompt Engineering           | Grounding rules for calendar/email data. Prevent inventing meetings. | Grounding rules for catalog data. Prevent inventing products.       | Distinguish stated vs assumed requirements. Refuse to generate from nothing. | Ground outputs in project files. Prevent fabricating milestones/statuses.             |
| Data Grounding               | Live API data (calendar events, emails)                              | CSV product catalog as grounding source                             | Transcript text as grounding source                                          | CSV files (timeline, risks, task board) as grounding source                           |
| Evaluation & Observability   | LangWatch or Langfuse. 12 test inputs.                               | LangWatch or Langfuse. 11 test inputs.                              | LangWatch or Langfuse. 12 test inputs.                                       | LangWatch or Langfuse. 12 test inputs.                                                |
| Tool Integration             | Google Calendar API + Gmail API via OAuth2                           | File upload nodes, CSV parsing                                      | File upload nodes, text parsing                                              | CSV parsing, optional Trello API                                                      |
| Fine-Tuning (optional)       | Train briefing or scheduler agent                                    | Train lead classifier or recommender                                | Train requirement extractor                                                  | Train planner or risk assessor                                                        |
| RAG (optional)               | Retrieve from calendar/email history                                 | Retrieve from product catalog                                       | Retrieve from PRD template + past PRDs                                       | Retrieve from project files                                                           |
| Human-in-the-Loop (optional) | Approve before creating events or sending emails                     | Approve before sending recommendations to customers                 | PM reviews extracted requirements before PRD generation                      | PM reviews generated plan before sharing with stakeholders                            |

## Project 1: CalendarMate

**Best for:** PMs/TPMs who spend significant time on meeting management, scheduling, and email triage.

You build an AI assistant that generates daily briefings from your calendar and email, schedules meetings via natural language, and summarizes your inbox. The challenge is connecting to live Google APIs (Calendar + Gmail) via OAuth2, which adds a setup step but makes the project feel real.

**Core capabilities:** Daily Briefing, Smart Scheduling, Email Summarization

**Extended (pick 1+):** Meeting Follow-Up, Conflict Resolver, Attendee Research, Weekly Digest

**Hallucination test:** Does the agent invent meetings? Does it schedule without checking availability? Does it fabricate email content?

**Technical note:** Requires Google OAuth2 credentials. If OAuth setup is challenging, you can use mock calendar/email data instead.

## Project 2: SalesGenie

**Best for:** PMs/TPMs involved in sales operations, customer inquiries, or product catalog management.

You build an AI assistant that reads inbound inquiry emails, classifies leads as Hot/Warm/Cold, recommends products from a catalog, and generates weekly sales summaries. Works entirely with file uploads (no OAuth needed). The core challenge is grounding recommendations in the actual product catalog.

**Core capabilities:** Lead Qualification, Product Recommendations, Weekly Sales Insights

**Extended (pick 1+):** Automated Follow-Up Drafting, CRM Auto-Update, Competitive Positioning

**Hallucination test:** Does the agent recommend products NOT in the catalog? Does it invent pricing? Does it classify vague emails as Hot?

**Technical note:** No API setup needed. Catalog CSV is the grounding source. Walnut Executive Desk is intentionally out of stock (test case T10).

## Project 3: PRD Genie

**Best for:** PMs/TPMs who write PRDs, work with engineering to scope features, or translate meeting discussions into structured documentation.

You build an AI assistant that ingests meeting transcripts and stakeholder notes, extracts requirements, generates structured PRDs following a template, and breaks them into epics and user stories. Has the richest sample data (5 transcripts including contradictory and incomplete ones) and tests whether your agent can distinguish stated facts from assumptions.

**Core capabilities:** Requirement Extraction, PRD Generation, Epic/User Story Breakdown

**Extended (pick 1+):** Gap Analysis, Scope Estimator, Version Comparison

**Hallucination test:** Does the agent invent requirements? Does it fill in 'TBD' items with assumptions? Does it generate a PRD from 'Meeting happened. Notes: none.'?

**Technical note:** No API setup needed. All inputs are text files. PRD template (prd\_template.md) defines expected output structure.

## Project 4: Mira

**Best for:** PMs/TPMs who manage project timelines, risk assessments, status reports, and milestone tracking.

You build an AI assistant that generates structured project plans from high-level descriptions, produces categorized risk assessments, and compiles weekly status reports from task board data. You are given a complete project dataset (ABCDE Ltd. AI Adoption Project) with a description, 6-month timeline, risk matrix, and a 25-task Kanban board export. The core challenge is grounding all outputs in this actual data.

**Core capabilities:** Project Plan Generation, Risk Assessment, Weekly Status Report

**Extended (pick 1+):** Milestone Alert System, Stakeholder Update Generator, Resource Allocation Advisor, Competitive Analysis

**Hallucination test:** Does the agent invent milestones not in the timeline? Does it fabricate task statuses? Does it generate generic risks instead of project-specific ones?

**Technical note:** No API setup needed if using CSV uploads. Optional Trello API integration. Task board has 1 deliberately Blocked task (T024).

# How to Choose

## By Role

| If you...                                           | Choose       |
|-----------------------------------------------------|--------------|
| Manage meetings and calendars daily                 | CalendarMate |
| Work in sales or deal with customer inquiries       | SalesGenie   |
| Write PRDs or translate meetings into requirements  | PRD Genie    |
| Manage project plans, timelines, and status reports | Mira         |

## By Concept You Want to Practice

| If you want to learn...                                              | Choose                         |
|----------------------------------------------------------------------|--------------------------------|
| Router/dispatcher orchestration with live API integration            | CalendarMate                   |
| Sequential pipeline with data grounding on structured files          | SalesGenie                     |
| Prompt engineering for distinguishing stated vs assumed requirements | PRD Genie                      |
| Multi-source CSV ingestion and data-grounded generation              | Mira                           |
| Working with Google OAuth2 and live APIs                             | CalendarMate                   |
| Working with file uploads only (fastest setup)                       | SalesGenie, PRD Genie, or Mira |

## By Setup Complexity

| Setup Level                          | Projects                    |
|--------------------------------------|-----------------------------|
| Fastest (file upload only, no OAuth) | SalesGenie, PRD Genie, Mira |
| Moderate (optional API integration)  | Mira (Trello API optional)  |
| Most setup (Google OAuth2 required)  | CalendarMate                |

## What Every Project Has in Common

- Multi-agent system design with documented orchestration pattern
- 3 core capabilities (required) + at least 1 extended capability
- Observability/tracing tool connected with traces visible
- Baseline test dataset (10-12 inputs) run and documented
- Architecture diagram and 1-2 page writeup
- Cost analysis and 3+ production metrics
- GitHub repository with workflow JSON, README, and screenshots
- Fine-tuning is optional but recognized in scoring

### Go beyond the baseline:

Once your core capabilities pass the test dataset, you are encouraged to add your own use cases, integrate additional APIs, improve prompts based on trace analysis, or attempt fine-tuning. The rubric rewards creativity and depth beyond the minimum requirements.